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Space Robot Dynamic Analysis of the Relative Orbital and Attitude Motion in the Close Range Rendezvous Phase and Grasping of a Target Space Vehicle

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Abstract

This article deals with the dynamic analysis of the close approach phase of a space robot to a target spacecraft aiming the grasping of a target space vehicle. The chaser robot shall berth by side of the target so as to have it inside its workspace envelop. The grasping operation shall be executed by inverse kinematics. The computer simulation of the equations of the dynamics (relative orbital and attitude equations of motion) yields the grasping conditions and such results are compared with the inverse kinematics grasping operation implemented in a lab experiment. In the experiment it is used two robot manipulators mounted on an air supported platform, constituting a floating robotic system that operates on a glass table. The experiment is wireless and this feature is obtained by using Wifi to implement the communication of the human operator with the robots and a platform containing its own air compressed cylinders. In addition to the WIFI the manipulator platforms include microcontroller, a set of valves to control the compressed air flow, and two compressed air cylinders. The actuators for rotational/translational motion operate on the base of compressed air. A sensor Kinect One sensor supported by computational vision is employed externally to the experiment. The use of such sensor aims to measure velocity, perform robot tracking as well as to obtain the X, Y, and Z positions of the manipulators' grippers at run time and 3D video scenes. The information is sent via WIFI, over the UDP protocol, to a computer with MATLAB® software, which writes the actual speed and positioning information and compares with the prescribed results (estimated in simulation). The experimental result obtained by inverse kinematics for the capture of the target is compared with the results of the computational simulations, aiming at the validation and improvement of the mathematical model simulated via MATLAB®. If the error does not meet the nominal specification the operator commands the robot system to continue the control loop. The compressed air autonomy is about 30 minutes for the platform floating and 1 minute for the grasping operation. The results are presented by using plots of the relative distance and velocity with respect to time, attitude synchronization with respect to time, and the robot-like spacecraft attitude motion during the safety-critical operations of grasping the target spacecraft. The direct kinematics obtained by the computational simulation is compared with the results of the inverse kinematics obtained via experiment.

Keywords: dynamic analysis, space robot, close range rendezvous

Symbols

$\boldsymbol{R}$ transformation matrix
$\boldsymbol{\omega}$, $\boldsymbol{v}$ angular and linear velocity vectors, respectively
$\widehat{\boldsymbol{Z}}$ Unit vector
$\boldsymbol{S}$ Position vector between joints
Lv Local Vertical, Local Horizontal frame
θ Joint angles
$\boldsymbol{J}(\Theta), \boldsymbol{J}_v(q), \boldsymbol{J}_{\boldsymbol{\omega}}(q)$ Jacobian matrices
q, Q Generalized coordinates and force, respectively
T kinetic energy
m mass
$i, n, k, 0, p, v, \omega, \sim$ Subscript/superscript for robot joints, coordinates, frames, platform, velocities, skew symmetric matrices
x, y, z, Cartesian coordinates
{ }, [ ] vector 3x1 and matrix notation respectively
s Complex variable

Acronyms

RVD/B Redezvouz and docking/Berthing
LVLH Local Vertical, Local Horizontal

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1. Introduction

This paper deals with the dynamics analysis of space system composed by, a chaser and a target spacecraft in the scenario of the rendezvous closed approach phase. The RVD refers to encounter of two spacecraft in orbit not necessarily followed by docking and/or berthing operations. In the case of this current paper the idea is to analyze the closed approach followed by berthing and capture of the target. The rendezvous docking/berthing is not a new subject. In fact since the sixties astronauts from NASA and the former Soviet Union had implemented those orbital operations. At that time those operations required always astronauts participation in the process of approximation and docking. Had not they implemented and tested such operations the man would not have landed on the Moon as planned by NASA. The plan to take the man to the Moon included an undocking and docking operation by the astronauts. They did it well and successfully. However the problem is still subject of several studies aiming mainly a complete autonomous operation of rendezvous and docking/berthing (RVD/B) with and without help from ground stations. One of the major risks of the RVD/B operations is collision and control of the connected spacecraft immediately after the docking or grasping end operation. Here is also another challenge for the attitude control designers: the spacecraft attitude control subsystem must be robust enough to control the attitude of connected system and to keep it stable in a prescribed attitude. In order to accomplish the proposed dynamics analysis the main tasks implemented are the mathematical modeling of a robotic spacecraft comprising a six-degrees of freedom robotic arm, a control design to track and control the relative motion during the close approach phase and the chaser spacecraft during the grasping operation of the target spacecraft.

1.1 The Close Range RVD/B

The phases of a RVD/B mission is presented and discussed in Fehse [1]. The main phases are launch, phasing, far range rendezvous, close range rendezvous and mating (docking or berthing). The close range rendezvous includes two sub phases, the closing and the final approach. During the closing the ACS must reduce the relative distance to target acquisition of final approach line. During the final approach the control must conduct the spacecraft safely to the capture conditions. In this phase any failure can result in the mission aborting or in collision. Accordingly the phase can be defined as safety-critical in agreement with the definitions used for the fault detection isolation and recovery planning. Regarding abort of such missions the main aspect to be considered is minimal distance from which it is not possible to abort the mission, resulting in collision. Another aspect that should be pointed out is that in this phase the relative navigation is implemented for computational simulations and analysis. The attitude control shall be designed so as to keep the robotic platform stable when the robot arm is acted by forces and torque to accomplish the task grasping or docking. The center of mass of the robotic spacecraft tends to move in reaction to those applied forces while the attitude tends to change in reaction to the applied torque in the arm joints.

2. Mathematical modeling of a robotic spacecraft with a robot arm mounted on its structure

2.1. Motion of the spacecraft about its center of mass

The mathematical model takes into account a platform where it is mounted the robotic manipulator. It is assumed that no external forces and torque affect the translational/rotational motion of the spacecraft during the close range rendezvous. Also the platform is considered rigid and the robot links as well. Figure 1 shows the robot configuration with the robot arm deployed. It represents a model of a floating robot in our Space Robot Lag and the mathematical model is used for the computer simulations by using MATLAB® software environment and for tests aiming the control algorithm validation.

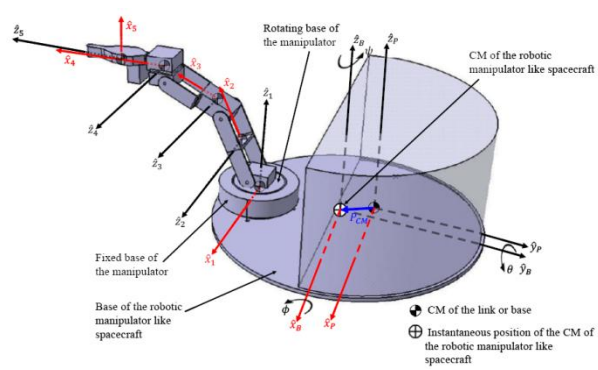

Fig. 1 Space Robotic model and frames defined for the sake of mathematical modeling

The derivation of the equations of motion is obtained by combining the approach described in Craig [2] to compute the velocities and the Lagrangian formulation as in Meirovitch [3]. To do this the manipulator is considered as chain of bodies, each one sharing the motion of the motion with the other (relative motion). Considering such structure, it is possible to compute the velocity of each link of the manipulator, starting from the base and going throughout the complete structure. In this way the velocity propagation for the robotic manipulator can be computed as in Craig [2]. By assuming the manipulator's links as rigid bodies with rotational and translation degrees-of-freedom the expression for the angular velocity can be written as

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$$ {}^{i}\boldsymbol{\omega}_{i+1} = {}^{i}\boldsymbol{\omega}_{i} + {}_{i+1}^{i}\boldsymbol{R}\dot{\theta}_{i+1}\,{}^{i+1}\hat{\boldsymbol{Z}}_{i+1} \tag{1} $$

Where

$$ \dot{\theta}_{i+1}\,{}^{i+1}\hat{\boldsymbol{Z}}_{i+1} = {}^{1+1}\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_{i=1} \end{bmatrix} \tag{2} $$

Then Eq.(1) can be written as

$$ {}^{i}\boldsymbol{\omega}_{i+1} = {}^{i}\boldsymbol{\omega}_{i} + {}_{i+1}^{i}\boldsymbol{R}\,{}^{1+1}\begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_{i+1} \end{bmatrix} \tag{3} $$

where ${}^{i}\boldsymbol{\omega}_{i+1}$ is the angular velocity of the frame i + 1 with respect to the frame i. Following this structure ${}^{i}\boldsymbol{\omega}_{i}$ is the angular velocity of frame i w.r.t. itself. ${}_{i+1}^{i}\boldsymbol{R}$ is the rotation matrix from frame i+1 to frame i. $\dot{\theta}_{i+1}\,{}^{i+1}\hat{\boldsymbol{Z}}_{i+1}$ is the anguar velocity of frame i+1 about $\hat{\boldsymbol{Z}}$ direction written w.r.t. itself. Eq.(3) is the matrix notation of Eq.(1), Similarly the linear velocity associated with the links can be written as

$$ {}^{i}\boldsymbol{v}_{i+1} = {}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{S}_{i+1} \tag{4} $$

Where ${}^{i}\boldsymbol{v}_{i+1}$ is the linear velocity of frame i+1 w.r.t. frame i, ${}^{i}\boldsymbol{v}_{i}$ is linear velocity of fram i w.r.t. itself. ${}^{i}\boldsymbol{S}_{i+1}$ is the vector position os the origin of link i+1 w.t.r. to the link i. This form to write the velocity for robots is called velocity propagation and is detailed in Craig' text book [2].

The pre multiplication Eq.(1) and Eq.(4) by ${}_{i}^{i+1}\boldsymbol{R}$ yields

$$ {}^{i+1}\boldsymbol{\omega}_{i+1} = {}_{i}^{i+1}R\ {}^{i}\boldsymbol{\omega}_{i} + {}^{i+1}\dot{\theta}\,{}^{i+1}\hat{\boldsymbol{Z}}_{i+1} \tag{5} $$

$$ {}^{i+1}\boldsymbol{v}_{i+1} = {}_{i}^{i+1}R(\ {}^{i}\boldsymbol{v}_{i} + {}^{i}\boldsymbol{\omega}_{i} \times {}^{i}\boldsymbol{S}_{i+1}) \tag{6} $$

By successive application of these equations from link to link it is possible to compute the rotational and the linear velocities up to ${}^{n}\boldsymbol{\omega}_{n}$ and ${}^{n}\boldsymbol{v}_{n}$. These equations can be written using the concept of Jacobian as [2]

$$ \boldsymbol{v} = \begin{bmatrix} v \\ \omega \end{bmatrix} = \boldsymbol{J}(\Theta)\dot{\Theta} \tag{7} $$

where $J(\Theta)$ is Jacobian matrix and $\dot{\Theta}$ is the vector of the joint velocities. As applied to robotic manipulators, $\boldsymbol{J}(\Theta)$ relates the motion of a desired point of the manipulator (the end effector, for example) with the joint angles, which means that $\boldsymbol{J}(\Theta)$ is a linear transformation from joints angles to Cartesian angular and linear rates of the manipulator end effector [4].

Eq.(7) can be written in two components, one for the linear $\boldsymbol{v}$ and another for the angular velocity $\boldsymbol{\omega}$ **as**

$$ \boldsymbol{v} = \boldsymbol{J}_{v}(q)\dot{\boldsymbol{q}} \tag{8} $$

$$ \boldsymbol{\omega} = \boldsymbol{J}_{\boldsymbol{\omega}}(q)\dot{\boldsymbol{q}} \tag{9} $$

Where $\boldsymbol{J}_{v}(q)$ and $\boldsymbol{J}_{\boldsymbol{\omega}}(q)$ are partial derivatives of the translational and rotational velocities, respectively, w.r.t. $\dot{\boldsymbol{q}}$. So $\boldsymbol{v}$ and $\boldsymbol{\omega}$ are a 3x1 vectors. Consistently J is 6 x i matrix. Using Eq.(8) the coordinate transformation from one frame to another is given by

$$ \boldsymbol{R}\boldsymbol{v} = \boldsymbol{R}\boldsymbol{J}_{v}(q)\dot{\boldsymbol{q}} \tag{10} $$

$$ \boldsymbol{R}\boldsymbol{\omega} = \boldsymbol{R}\boldsymbol{J}_{\boldsymbol{\omega}}(q)\dot{\boldsymbol{q}} \tag{11} $$

where $\boldsymbol{R}$ is the rotation matrix. In this paper there is a platform to which a 5 degrees-of-freedom robot manipulator is attached (see figure 1). The transformation matrices relating the several frames with the LVLH frame can be written as

$$ \begin{aligned} {}_{p}^{Lv}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R} \\ {}_{1}^{v}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R}\,{}_{1}^{p}\boldsymbol{R} \\ {}_{2}^{v}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R}\,{}_{1}^{p}\boldsymbol{R}\,{}_{2}^{1}\boldsymbol{R} \\ {}_{3}^{v}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R}\,{}_{1}^{p}\boldsymbol{R}\,{}_{2}^{1}\boldsymbol{R}\,{}_{3}^{2}\boldsymbol{R} \\ {}_{3}^{v}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R}\,{}_{1}^{p}\boldsymbol{R}\,{}_{2}^{1}\boldsymbol{R}\,{}_{3}^{2}\boldsymbol{R}\,{}_{4}^{3}\boldsymbol{R} \\ {}_{3}^{v}\boldsymbol{R} &= {}_{b}^{Lv}\boldsymbol{R}\,{}_{p}^{b}\boldsymbol{R}\,{}_{1}^{p}\boldsymbol{R}\,{}_{2}^{1}\boldsymbol{R}\,{}_{3}^{2}\boldsymbol{R}\,{}_{4}^{3}\boldsymbol{R}\,{}_{5}^{4}\boldsymbol{R} \end{aligned} \tag{12} $$

where Lv stand for LVLH frame, p means platform (the manipulator platform) 1,2...5 are the frames defined at frame each joint. Considering the complete robotic manipulator like-spacecraft, the transformation from the platform frame p to LVLH frame is:

$$ {}_{p}^{Lv}\boldsymbol{R} = \begin{bmatrix} c\psi c\theta & c\psi s\theta s\phi - s\psi c\phi & c\psi s\theta c\phi + s\psi s\phi \\ s\psi c\theta & s\psi s\theta s\phi + c\psi c\phi & s\psi s\theta c\phi - c\psi s\phi \\ -s\theta & c\theta s\phi & c\theta c\phi \end{bmatrix} \tag{12} $$

where the matrix elements are the direction cosines in the attitude angles for se Euler sequence of rotations from Lv , $R_x(\phi)R_y(\theta)R_z(\psi)$.

$$ {}_{1}^{p}\boldsymbol{R} = R_z(\theta_1) = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 \\ s\theta_1 & c\theta_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{13} $$

$$ {}_{2}^{1}\boldsymbol{R} = \boldsymbol{R}_z(90^o)\boldsymbol{R}_x(90^2)\boldsymbol{R}_z(\theta_2) = \begin{bmatrix} 0 & 0 & 1 \\ c\theta_2 & -s\theta_2 & 0 \\ s\theta_2 & c\theta_2 & 1 \end{bmatrix} \tag{14} $$

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$$ {}^{2}_{3}\boldsymbol{R} = \boldsymbol{R}_z(\theta_3) = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 \\ s\theta_3 & c\theta_3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{15} $$

$$ {}^{3}_{4}\boldsymbol{R} = \boldsymbol{R}_z(\theta_4) = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 \\ s\theta_4 & c\theta_4 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{16} $$

$$ {}^{4}_{5}\boldsymbol{R} = \boldsymbol{R}_z(-90^o)\boldsymbol{R}_x(-90^2)\boldsymbol{R}_z(\theta_2) = \begin{bmatrix} 0 & 0 & 1 \\ -c\theta_5 & s\theta_5 & 0 \\ -s\theta_5 & -c\theta_5 & 1 \end{bmatrix} \tag{17} $$

$$ {}^{B}_{p}R = {}^{p}_{1}\boldsymbol{R}\,{}^{1}_{2}\boldsymbol{R}\,{}^{2}_{3}\boldsymbol{R}\,{}^{3}_{4}\boldsymbol{R}\,{}^{4}_{5}\boldsymbol{R} \tag{18} $$

By inserting these transformations into Eqs.(5-6) these yields,

$$ \{{}^{p}_{p}v\} = [{}^{p}_{b}R][\widetilde{\omega}_p]\{r_p\} + \{\dot{r}_p\} \tag{19} $$

$$ \{{}^{Lv}_{i}v\} = \{\dot{x}_{0i} \quad \dot{y}_{0i} \quad \dot{z}_{0i}\}^T \tag{20} $$

where $[\widetilde{\omega}_p]$ is a skew symmetric matrix in the components of ω. The subscript 0 means frame 0. $\{\ \}$ means a 3X 1vector

where $i = 1,2,..,5\ (joints)$. Similarly

$$ \{{}^{b}\omega_b\} = [{}^{b}_{Lv}R][\Theta]\{\omega\} - n[{}^{b}_{Lv}R]\{f(\theta,\phi,\psi\} \tag{21} $$

$$ {}^{2}_{2}\omega = [{}^{2}_{1}R]\{{}^{1}\omega_1\} + \{0 \quad 0 \quad \dot{\theta}_2\}^T \tag{22} $$

$$ {}^{3}_{3}\omega = [{}^{3}_{2}R]\{{}^{2}\omega_2\} + \{0 \quad 0 \quad \dot{\theta}_3\}^T \tag{23} $$

$$ {}^{4}_{4}\omega = [{}^{4}_{3}R]\{{}^{3}\omega_3\} + \{0 \quad 0 \quad \dot{\theta}_4\}^T \tag{24} $$

$$ {}^{5}_{5}\omega = [{}^{5}_{4}R]\{{}^{4}\omega_4\} + \{0 \quad 0 \quad \dot{\theta}_5\}^T \tag{25} $$

These equations are used to derive the kinetic energy to derive de equations of motion by using the Lagrangian formulation.

The kinetic energy can be written as a summation of the kinetic energy of the robotic spacecraft parts, i.e., the kinetic energy of the platform and the kinetic energy of each link of the robot arm. Then the kinetic energy of each link of the robot arm is:

$$ T = Tp + \sum_{i=1}^{5} T_i = \frac{1}{2}\{\omega\}^T[I_T]\{\omega\} + \frac{1}{2}m_i\left\{v_{c_p}\right\}^T\left\{v_{c_p}\right\} + \frac{1}{2}\sum_{i=1}^{5}\{{}^{Lv}_{i}\omega_i\}^T[{}^{i}_{i}I_i]\{{}^{Lv}\omega_i\} + \frac{1}{2}\sum_{1}^{5} m_i v_{c_i}^T v_{c_i} $$

Where v_{c_i} and ${}^{i}\omega_i$ are functions of the generalized coordinate coordinates and velocities, $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$, respectively. Note that each link angular velocity includes also the platform rotation motion. So after a hard algebraic manipulation it is possible to write the inertia matrix including all the contribution on the links moment of inertia together with [*I*].

Where *p* means platform and *i=1…5* refers to the 5 joints of the robot arm.

$$ T = \frac{1}{2}m_p {v_{c_p}}^T v_{c_p} + {}^{p}\omega_p\,{}^{Lv}_{p}I\,{}^{p}\omega_p + \sum_{i=1}^{5}\left(m_i {v_{c_i}}^T v_{c_i} + {}^{i}\omega_i^{T\,c_i}I\ {}^{i}\omega_i\right) \tag{26} $$

The term in summation refers to each link of the robot arm.

Taking the velocities equations, Eq. (19) up to Eq. (25), calculating their Jacobians given by Eq. (10) and Eq. (11), rotating all of them to the same reference frame by the use of the derived rotation matrices, Eqs. (12), and replacing them into Eq. (26), it is obtained kinetic energy for each component of the robotic manipulator like-spacecraft as

$$ T = \frac{1}{2}\{\dot{q}\}^T\left(m_p\left[J_{v_p}\right]^T\left[J_{v_p}\right] + \left[J_{\omega_p}\right]^T\left[{}^{Lv}_{p}I\right]\left[J_{\omega_p}\right] + \sum_{i=1}^{5}\left[J_{v_i}\right]^T + [{}^{Lv}_{i}I_i]^T[{}^{Lv}_{i}I]\left[J_{\omega_i}\right]\right)\{\dot{q}\} \tag{27} $$

Where

$$ [{}^{Lv}I_p] = [{}^{Lv}_{p}R]^T[\ {}^{p}I_p][{}^{Lv}_{p}R] $$

$$ {}^{v}I_i = [{}^{Lv}_{i}R]^T[\ {}^{i}I_i][{}^{Lv}_{i}R] $$

In more compact form:

$$ T = \tfrac{1}{2}\{\dot{q}\}^T\left[J_{\omega_p}\right][D]\{\dot{q}\}^T \tag{28} $$

where D is the matrix coefficient of $\{\dot{q}\}$ in Eq.(27).

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Note that the transformation matrices do not appear for the linear velocities for those velocities are written straightforward in the Lv frame.

The appropriate Lagrange's formula to obtain the equations of motion is

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right) - \left(\frac{\partial T}{\partial q_k}\right) + \frac{\partial F}{\partial \dot{q}_k} = Q_k \tag{29}$$

Where F is a Rayleigh Ritz [3] dissipative function given by

$$F = \frac{1}{2}\sum_{r=1}^{5} c_r\, \dot{q}_r^2 \tag{30}$$

This formulation account for Eq.(30) allows the inclusion of the dissipative effects in the joints.

The derivation of the equations of motion obtained by application of the Lagrange's formula, Eq.(29), yields

$$\begin{gathered}
\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{q}_k}\right) = \frac{d}{dt}([D]\{\dot{q}_k\}) = [D]\{\ddot{q}_k\} + [\dot{D}]\{\dot{q}_k\} \\
\left(\frac{\partial T}{\partial q_k}\right) = \frac{1}{2}\{\dot{q}_k\}^T \frac{\partial [D]}{\partial q_k}\{\dot{q}_k\} \\
\frac{\partial F}{\partial \dot{q}_k} = c_r \dot{q}_k \\
[D]\{\ddot{q}_k\} + [\dot{D}]\{\dot{q}_k\} - \tfrac{1}{2}\{\dot{q}_k\}^T \tfrac{\partial [D]}{\partial q}\{\dot{q}_k\} = Q_k
\end{gathered} \tag{31}$$

Q_k can be any other type of forces that are not derived from a potential field. In this problem Q_k accounts for the control generalized forces/torque. No external environmental perturbation is considered in this study.

The Lagrangian formulation yields a set of second order differential equations. There 20 second order equations for the robot joints (translation and rotation), and three for the satellite attitude motion (roll, pith, yaw). The state vector is 2n dimensional. Then the total of state variables is 2n, i.e., 46 variables. The 23 second order differential equations obtained via Lagrangian formulation for this problem does not include the orbital relative motion (Hills equations). The mathematical modeling for the relative orbital motion is approached next.

2.2. The equations of motion in the target reference frame.

Whereas during the phasing the maneuvers are based on absolute navigation, for orbital operations the far and close range rendezvous maneuvers are based on the relative navigation. It means that the motion under consideration is the relative motion between the center of mass of both spacecrafts. Figure 2 illustrates the relative position between both spacecrafts. The time derivative of the relative position yields the translational velocity between one spacecraft w.r.t. another.

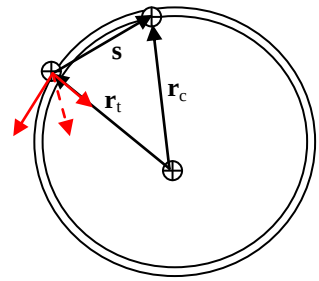

Fig. 2. Chaser and Target Orbital Configuration

Usually the frame defined in the center of mass of the target (or eventually in the center of mass of the chaser) points the axes *z* toward the center of the Earth, *x* toward the orbital motion, and *y* toward the orbit normal completing the right handed frame. This type of frame is named Local Vertical, Local Horizontal, LVLH. For rendezvous and docking operations *z* is called *z*-bar, *y* is called H-bar, and *x*, *x*-bar. This frame is a heritage from aeronautics and it is associated with roll (*x*), pitch (*y*), and yaw (*z*) motions. In general the frame is defined in the target. The idea is that the target be an alive spacecraft or an active space station, so as to exchange information with the o chaser to collaborate with the RVD/B operation. If the target is non collaborative (a dead satellite or another object, for example) there is no reason to define the LVLH frame in the target.

There are some safety-critical issues related to the navigation and control requirements for safety docking and berthing. Measurements accuracies in the final approach are approximately 1% (about a few cm in lateral position and 0.10 (deg/s). However this is not a closed requirement since the accuracies depend on the reception range. For berthing those requirements are less stringent for relative position but much more stringent for lateral and attitude

The equation for the translational motion can be derived by using the Newton approach as. The detailed derivation of these equations and can be seen in Fehse [1], appendix A. The linear equations can be written as

$$\ddot{x} - 2\omega\dot{z} = \frac{1}{m_c}F_x \tag{32}$$

$$\ddot{y} + \omega^2 y = \frac{1}{m_c}F_y \tag{33}$$

$$\ddot{z} + 2\omega y - 3\omega z = \frac{1}{m_c}F_z \tag{34}$$

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This system of linear equations is a general system valid for an arbitrarily relative trajectory between a chaser and a target spacecraft.

3. Control Techniques

For good performance the method must be used carefully by tuning the gains. A generic block diagram illustrating the PID control technique is shown in figure 2.

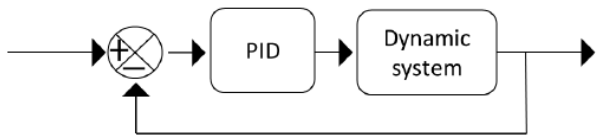

Fig. 3. PID block diagram

The transfer function associated with this technique can be written as

$$TF = k_p\left(1 + \frac{1}{T_i s} + T_d s\right)$$

where TF means transfer function of the controller. K_p is the proportional gain and T_i and T_d are the integral and the derivative time constants, respectively. The meaning of the gains K_p, K_i, and K_d are shown in table 1.

This method is adopted here for the control of the manipulator like-spacecraft attitude, joints motion and the relative motion as well. In the next the discussion on the PID is extended to the effect of the gains as shown in table 1.

Table1. Meaning of the PID controllers

Gains	Rise Time	Overshoot	Settling time	error
K_p	Decrease	Increase	Small change	Decrease
K_i	Decrease	Increase	Increase	Eliminate
K_d	Small change	Decrease	Decrease	Small change

The correlations in table 1 may not be exactly accurate. The reason is that changing one of those parameters can change the effect of the other two. The tuning of the parameter allows better PID control performance.

The equations of motion subjected to the computational simulations refer to the coupled attitude equations with the robot arm equations of motion. The robot arm equations are associated with the degrees-of-freedom of the robotic manipulator.

Table 2 shows the PID gains used in the computational simulation of the system equations. The state vector is 15x1.

Table 2. Parameter used in the computer simulations

	Attitude	K_p	K_i	K_d
Robotic spacecraft	Roll	100	5	20
	Pitch	100	5	20
	Yaw	100	5	20
Manipulator mounted on the robotic spacecraft	Joint 1	100	5	25
	Joint 2	70	5	15
	Joint 3	70	5	15
	Joint 4	70	5	15
	Joint 5	70	5	15

The state system equations are large and complicated because there are, in addition to the attitude equations, 5 equations for angular motion of the links and 4 other equations for the joint translations. The equations for robot degrees-of-freedom involve several transformation matrices products and the resulting equations for the links are very complicated. The state vector is defined as composed by the n position variables and n rates, yielding $2n$ first order equations in an appropriate form for numerical integration. In this case there 10 equations associated with the manipulator links and 3 more equations related to the attitude motion, resulting in 13 second order differential equations and 26 first order equations in the state variables.

4. Computer simulations

The computer simulations were implemented in the MATLAB® software and Simulink environment software for two sets of equations, the rotational equations of the robot spacecraft like-robotic manipulator

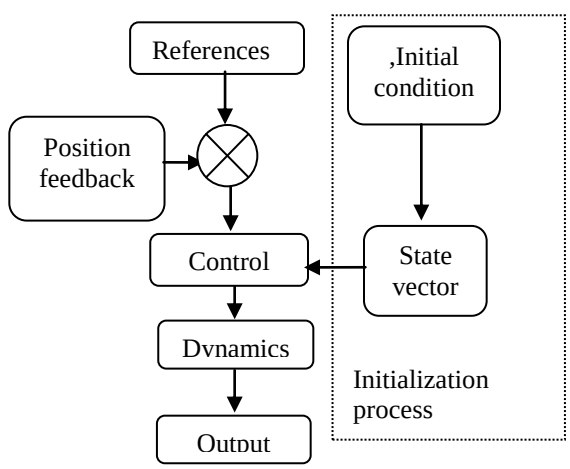

Fig. 4. Diagram of the numerical solution process.

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5. Experiment and Control Algorithm Tests

This work includes a test workbench to validate control algorithms used in computational simulations of orbital operations of rendezvous and docking/berthing and for on-orbit servicing as well. The experiment comprises a balanced glass table, on which two air supported six degrees-of-freedom robot manipulators floats on a thin layer of air. The air supported system aims the reduction or elimination of any significant friction forces during the rotational/translational motion of the manipulators. The orbital environment is characterized by microgravity where orbiting spacecraft and objects experience lack of weigh and floats. Such characteristics of the experiment are appropriate to test orbital and attitude rendezvous maneuvers and on-orbit servicing maneuvers as well. Problems associated with those operations are associated to the action and reaction Newton law.

The robotic platform comprises three small circular supports with micro holes where the air flows under pressure so as to keep the platform suspended and free to slide even when a weak force acts on it. The manipulator motion is restricted to a planar motion in spite of the arm joints can move in a 3-D space. The most important feature is that it is possible to test motion of the robot platform in response to the commanded actions of the control to the arm while the robot manipulator operates in space. This problem requires control actions to keep the platform stable so that the robot manipulator can accomplish successfully its work in space. Figure 5 illustrates the robot manipulator.

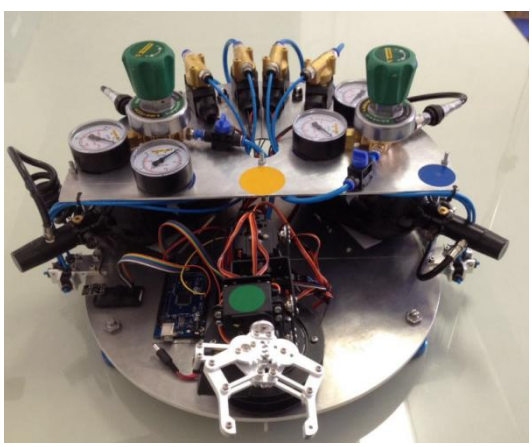

Fig. 5. Air supported robot manipulator on a balanced glass table.

The robot does not carry any type of sensor on structure. Its movements, position and direction are obtained by a system of capture and processing of images using the sensor Kinect One, manufactured by Microsoft. Figure 6 shows the positioning of Kinect over the plane of the glass table.

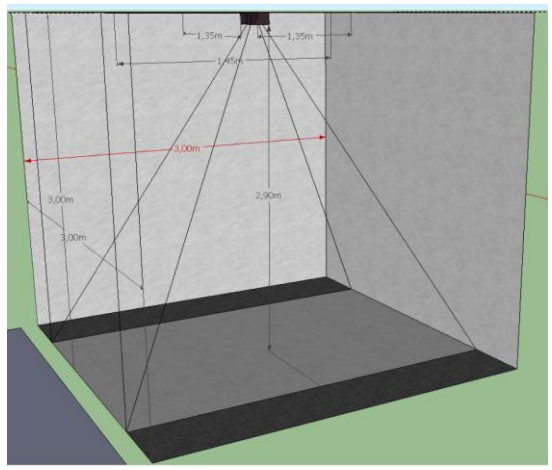

Fig. 6. Kinect Tracking System

This sensor is composed of a set of cameras that captures the image (RGB) and Depth (HD). The integration of depth information and color data results in a cloud of colored dots containing about 300,000 dots in each frame. By recording the RGB and depth images consecutively, a higher density of points can be obtained with acceptable accuracy [5,6], thus allowing a complete cloud of points to be created from an internal, possibly real-time environment [7]

With this capture, the data is sent to a computer that analyzes the images and produces a record of position, direction and speed of each of the robots in the experiment. The location operation is aided by round and colored labels that aid the process of recognizing the robot and its positioning. This information is sent to a second computer that controls the experiment. With the information received the computer adjusts the commands to be sent to the robots. Each robot receives commands through a WIFI connection via UDP protocol. These implemented commands and associated results are collected by the capture module (Figure 7).

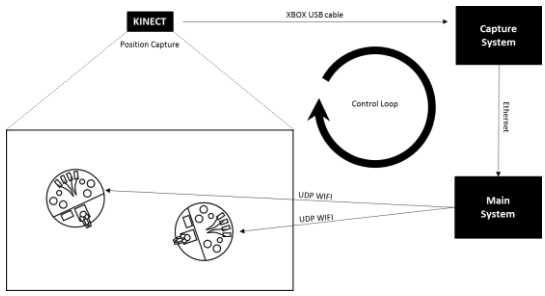

Fig. 7. Control loop between the manipulators

The capture system must have a set of functionalities so that it can operate properly in order to provide

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sufficiently precise information aiming the control of the movements of the robots in an effective way. The functionalities are

- Capability to identify colored marks placed on the table, on the robots and on the articulated arms (Figure 8).

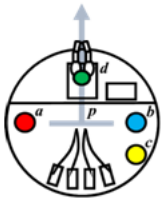

Fig. 8. Collored Marks to be tracked by the Kinect

- Calculate the position and direction of the database in relation to the glass table. The position of the robot is midpoint between a and b in figure 6,
- Determine the position of the robot direction defined by the vector $\mathbf{u} = (c, b)$
- Identify the position d of the robotic arm of each robot. The hand of the robot has a colored mark whose position is compared with the position of the robot and its respective direction to identify its position relative to the base of the robot
- Send through the UDP protocol the packets of the sensor signals that capture signals of luminosity (lumens), light intensity (luxs) and ultraviolet (UV) rays of the environment. The variation of these signals alters the accuracy of the coordinates. Therefore, these signals are captured and sent to the control system.
- Send via the UDP protocol the position data packets to the control system (Ethernet connection).

6. Computer Simulations Results

The computer simulations were implemented in the MATLAB® and Simulink Software environment. Two different control techniques have been used. The PID technique has been used to control rotational equations of the platform containing the robotic manipulator. The equations of dynamics is compact form is given by the last of Eqs.(31), written in state space variables (2n equations). The orbital relative motion of the target spacecraft w.r.t. the chaser has been integrated by using the Hill equations, Eq.(32-34).

The focus in this simulation is to actuate on the manipulator so that it operates in space toward the target. Figure 9 show the case in which the controllers actuate in the joint to move the arm links and the expected control efforts to keep the on going operation.

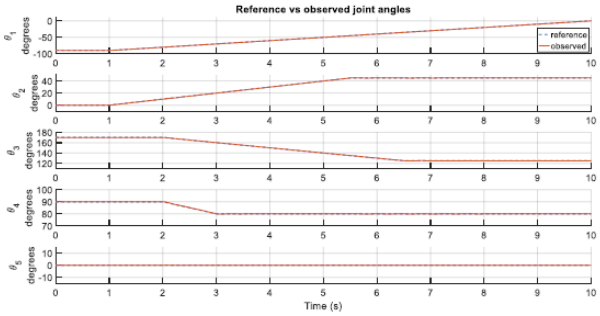

Fig. 9. Reference and the observed joint angles considering the control of the attitude and the manipulator in movement.

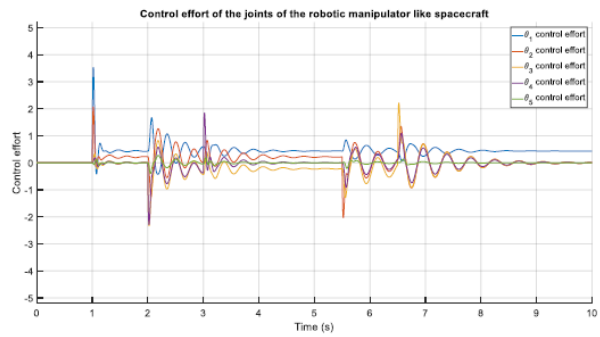

Fig. 10. Control effort to implement the case shown in figure 9

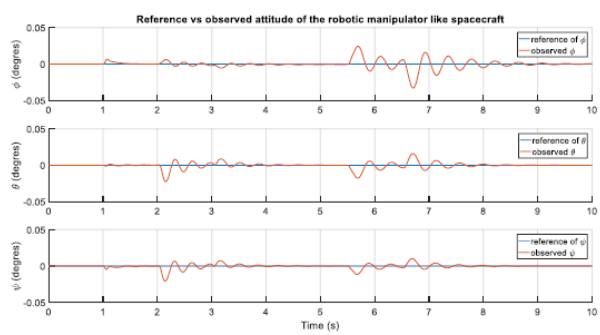

Fig. 11. Disturbance caused in the spacecraft attitude due to the operation of the robot arm.

Figure11 exhibits the disturbance caused on the attitude while the arm is being commanded for operations. Control actions eliminate the disturbance effects keeping the platform stable. Figure 12 shows the control effort to eliminate the perturbation shown in figure 9 and keep the spacecraft under control.

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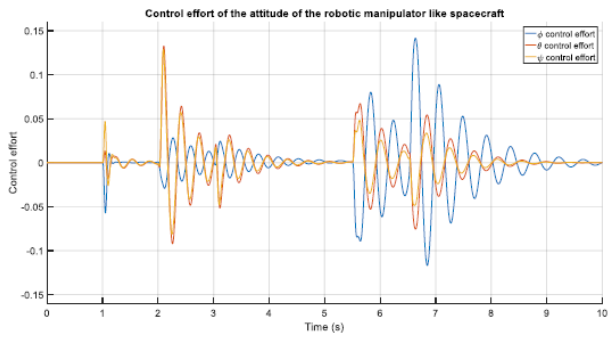

Fig. 12. Control effort to stabilize the spacecraft while the robot manipulator is operating.

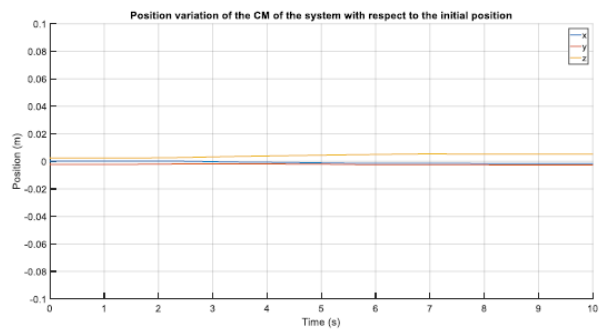

Fig. 13 Changes in the Center of Mass when the Arm is Operating.

The next figures show the simulation and control of the orbital relative motion up to the final approach.

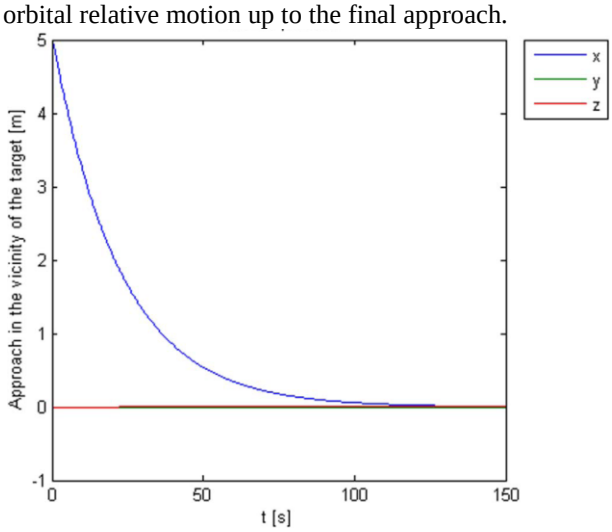

Fig. 14 Positioning close to the target

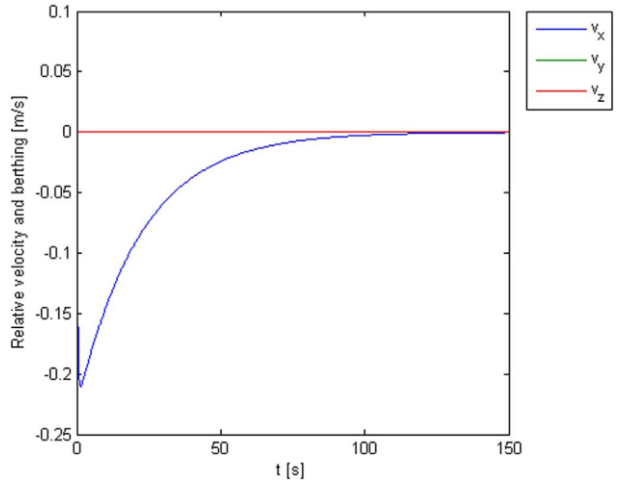

Fig. 15. Relative velocity - final approach to the target

6. Conclusions

This paper presents the mathematical modeling for a spacecraft like-manipulator. It is analyzed the impact of arm movement on the attitude of the spacecraft. The control effort is analyzed two scenarios, one to keep the attitude stable while operating the manipulator and the other the control effort to move the joints of the robot toward the target. The relative orbital motion is analyzed for the berthing close to the target considering the relative distance and the relative velocity. The paper presents the test workbench in which the control algorithm is being tested. The main contributions of this work are the analysis of the problem of stabilizing the robotic platform while the manipulator operates and the usage of computational vision to implement the simulations related to the lab experiments. The lab test confirms the efficiency of PID controller used in the computer simulations of the Hills equations (refers to figures 14 and 15.

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